

LESSON 7-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Comparing Numbers on a Number Line**

Maryland Standard 6.NS.C.7a · Learning target – I can compare two rational numbers using their positions on a number line.

REMEMBER

- Numbers get **greater to the right** and **less to the left**.
- Any positive number is greater than any negative number.
- $-2 > -8$ because -2 is farther right. $<$ means is less than; $>$ means is greater than.

WORKED EXAMPLE

Compare -3.5 and -1.5 . Use $<$ or $>$.

Step 1 -1.5 is to the right of -3.5 on the number line.

Step 2 Right means greater.

Answer $-1.5 > -3.5$

WORD BANK

greater than ($>$) – farther right on the number line **compare** – decide which number is greater

less than ($<$) – farther left on the number line

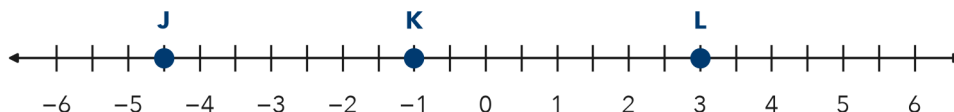
PRACTICE

show your work · label units

1. Use the number line. Write $<$ or $>$ in each blank.

Step 1 Find the number for J, K, and L.

Step 2 The point farther right is greater.



a. J $\underline{\hspace{1cm}}$ K

b. L $\underline{\hspace{1cm}}$ K

c. J $\underline{\hspace{1cm}}$ L

2. Write $<$ or $>$ in each blank.

Step 1 Picture both numbers on a number line.

Step 2 Write $>$ if the first number is farther right. Write $<$ if it is farther left.

a. $-8 \underline{\hspace{1cm}}$ -3

b. $0 \underline{\hspace{1cm}}$ -6

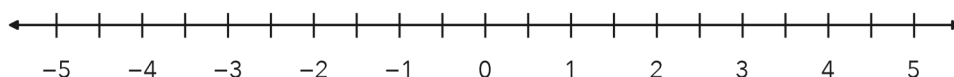
c. $-2.5 \underline{\hspace{1cm}}$ -2.05

d. $4 \underline{\hspace{1cm}}$ -10

e. $-1/2 \underline{\hspace{1cm}}$ $-3/4$

f. $-7 \underline{\hspace{1cm}}$ -7.1

3. Plot -3.5 and -0.5 on the number line. Then write a true comparison using $<$. _____



4. Golf. In golf, the **lower** score wins. Tanya scored -3 . Malik scored -5 .

a. Compare the scores using $<$ or $>$. _____

b. Who won? _____

5. Cold Days. Monday's low was -7 degrees. Tuesday's low was -2 degrees. Which day was colder? Write a comparison using $<$ or $>$.

Answer _____

6. Reasoning. Is $-6 < |-6|$ true? Explain how you know.

Frame $|-6| = \underline{\hspace{1cm}}$. -6 is to the $\underline{\hspace{1cm}}$ of 6 on the number line, so $-6 < |-6|$ is $\underline{\hspace{1cm}}$.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Ask someone at home: which is colder, -4 degrees or -9 degrees? Why?

LESSON 7-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Ordering Integers and Rational Numbers**

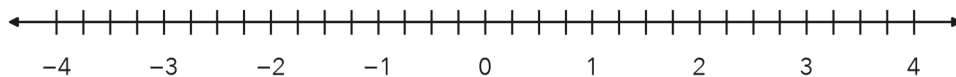
Maryland Standard 6.NS.C.7b • Learning target – I can order integers and rational numbers from least to greatest.

REMEMBER

- Place the numbers on a number line. **Least** is farthest left; **greatest** is farthest right.
- Change fractions to decimals to compare: $-3/4 = -0.75$.
- For negative numbers, the one farther from 0 is **less**.

WORKED EXAMPLEOrder -1.2 , 0.5 , -2 , and $-1/2$ from least to greatest.**Step 1** Write $-1/2$ as -0.5 .**Step 2** Read the number line from left to right.**Answer** $-2, -1.2, -1/2, 0.5$ **WORD BANK****order** – put numbers in a row**least** – farthest left**greatest** – farthest right**deepest** – the least (most negative) number**PRACTICE***show your work • label units*

1. Plot each number. Then write them in order from least to greatest: 1.5 , -2.5 , $-3/4$, $2\ 1/4$, -3.25

Step 1 Change $-3/4$ and $2\ 1/4$ to decimals.**Step 2** Plot each number.**Step 3** Read the points from left to right.**Order** _____

2. Order each set from least to greatest.

Step 1 Put the negative numbers first. The one farthest from 0 goes first.**Step 2** Then write 0 and the positive numbers.**a.** $-6, 4, -9, 0$ _____**b.** $-0.3, -3, -0.03$ _____

3. Circle the greatest number in each set.

a. $-5, -2, -8$ **b.** $-1.4, -1.04, -1.44$ **c.** $-2/3, -1/2, -3/4$ **d.** $0, -0.1, -10$

4. **Ocean Animals.** A shark swims at -45.5 m. A turtle is at -12 m. A dolphin is at -30.25 m. A jellyfish is at -8.75 m. List them from **deepest** to **closest to the surface**.

Answer _____

- 5. Find the Error.** Leo ordered $-\frac{1}{2}$, -0.4 , and -0.6 from least to greatest as -0.4 , $-\frac{1}{2}$, -0.6 . Explain his mistake and write the correct order.

Frame Leo is wrong because for negative numbers, farther from 0 is _____. The correct order is _____.

TEST PRACTICE

choose the best answer

- 6.** Which list is ordered from least to greatest?

A $-5, -8, 0, 3$

B $-8, -5, 0, 3$

C $0, -5, -8, 3$

D $3, 0, -5, -8$

- 7.** Which number is greatest?

A -2.7

B -2.07

C -2.77

D -2.2

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME List the low temperatures for three cold days with someone at home. Order them from coldest to warmest.